

KENORA, ON, Canada

WMO: 718500

Lat: 49.791N

Lon: 94.361W

Elev: 410

StdP: 96.50

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.2	-28.7	-35.4	0.1	-30.9	-33.1	0.2	-28.3	8.6	-12.3	8.1	-12.5	3.4	310	0.639

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.8	28.6	20.0	27.0	19.0	25.5	18.1	21.7	26.4	20.5	25.1	19.5	23.8	4.8	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.1	15.5	24.5	18.9	14.4	23.2	17.8	13.4	21.9	65.5	26.3	61.3	25.2	57.6	23.8	25.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.4	7.7	6.8	-34.1	31.6	2.7	1.6	-36.1	32.7	-37.6	33.7	-39.1	34.6	-41.1	35.7		
(5)			-34.3	23.4	2.6	1.2	-36.2	24.3	-37.7	25.0	-39.2	25.7	-41.1	26.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	3.3	-15.8	-13.0	-5.0	3.4	10.7	17.0	19.8	18.6	13.4	5.5	-3.5	-12.2	(6)
(7)		DBStd	13.69	7.87	7.41	7.48	5.94	5.09	3.93	3.19	3.31	4.61	4.91	6.33	7.40	(7)
(8)		HDD10.0	3441	799	644	467	207	54	2	0	0	18	157	407	687	(8)
(9)		HDD18.3	5629	1057	877	724	447	239	68	21	37	158	398	657	945	(9)
(10)		CDD10.0	1010	0	0	1	10	77	213	303	267	122	17	0	0	(10)
(11)		CDD18.3	156	0	0	0	0	4	29	66	46	11	1	0	0	(11)
(12)		CDH23.3	934	0	0	0	1	32	196	393	259	53	1	0	0	(12)
(13)		CDH26.7	169	0	0	0	0	3	37	75	48	6	0	0	0	(13)
(14)	Wind	WSAvg	3.8	3.7	3.7	3.8	3.9	3.9	3.7	3.5	3.5	3.8	4.0	3.9	3.7	(14)
(15)	Precipitation	PrecAvg	674	29	23	35	40	59	102	93	92	76	51	43	31	(15)
(16)		PrecMax	1054	85	72	68	95	133	257	226	240	187	176	107	53	(16)
(17)		PrecMin	394	7	2	8	0	9	24	21	24	11	6	10	16	(17)
(18)		PrecStd	150	16	13	17	26	31	49	45	51	47	34	23	12	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.3	5.0	14.7	20.9	26.6	30.6	31.0	30.2	27.5	21.6	12.8	3.2	(19)
(20)			MCWB	0.9	2.9	9.1	11.4	16.2	20.8	21.9	20.9	19.8	15.2	9.1	1.4	(20)
(21)		2%	DB	-0.1	2.3	9.8	17.5	23.7	27.2	28.6	27.9	24.8	17.9	9.0	0.7	(21)
(22)			MCWB	-0.7	0.5	5.2	9.1	14.5	18.3	20.2	19.8	18.1	13.0	6.3	-0.4	(22)
(23)		5%	DB	-2.4	0.1	6.8	14.7	21.4	25.5	27.1	26.1	22.5	15.1	6.6	-0.6	(23)
(24)			MCWB	-3.1	-1.0	3.4	7.4	13.1	17.3	19.2	18.9	16.6	11.1	4.3	-1.4	(24)
(25)		10%	DB	-4.7	-2.0	4.3	12.4	19.0	23.8	25.6	24.5	20.4	12.7	4.7	-2.1	(25)
(26)			MCWB	-5.4	-3.3	1.6	6.3	11.8	16.7	18.6	18.0	15.7	9.5	3.0	-2.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.3	3.2	10.4	12.1	18.6	22.2	23.2	22.7	21.2	16.1	9.5	1.7	(27)
(28)			MCDWB	2.2	4.5	13.1	19.2	23.0	27.2	28.4	27.5	25.4	19.0	11.6	2.9	(28)
(29)		2%	WB	-0.8	0.7	6.1	10.0	16.3	20.2	21.9	21.2	19.4	13.7	6.8	-0.1	(29)
(30)			MCDWB	-0.2	2.0	9.4	15.6	21.1	25.9	26.6	25.9	22.8	17.5	8.4	0.5	(30)
(31)		5%	WB	-3.1	-0.9	3.7	8.3	14.4	18.8	20.8	20.0	17.7	11.6	4.7	-1.4	(31)
(32)			MCDWB	-2.5	0.2	6.4	13.7	18.8	23.2	25.1	24.7	21.1	14.5	6.4	-0.7	(32)
(33)		10%	WB	-5.3	-3.3	1.7	6.7	12.9	17.5	19.7	19.0	16.3	9.5	2.8	-2.8	(33)
(34)			MCDWB	-4.8	-2.1	4.0	11.9	18.0	21.9	23.9	23.1	19.6	12.3	4.5	-2.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.1	9.1	9.4	9.7	9.8	9.0	8.8	8.6	8.1	6.5	5.8	6.8	(35)
(36)			MCDBR	8.0	8.5	10.2	12.6	12.3	11.0	10.1	10.3	10.5	10.1	7.6	7.0	(36)
(37)			MCWBR	7.5	7.1	6.9	7.3	6.2	5.0	4.7	4.9	5.6	6.4	5.6	6.3	(37)
(38)		5% WB	MCDBR	7.9	8.2	9.8	11.6	10.4	9.4	9.1	9.4	9.2	9.3	7.0	6.7	(38)
(39)			MCWBR	7.5	7.0	7.0	7.3	6.2	5.4	5.2	5.2	5.8	6.5	5.5	6.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.246	0.254	0.286	0.325	0.343	0.366	0.377	0.363	0.343	0.298	0.267	0.239	(40)	
(41)		taud	2.299	2.290	2.346	2.439	2.468	2.426	2.415	2.482	2.515	2.599	2.473	2.327	(41)	
(42)		Ebn at Noon	817	904	929	923	916	892	876	874	858	851	795	775	(42)	
(43)		Edh at Noon	67	87	100	103	105	110	110	98	86	66	57	56	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.24	2.40	3.62	4.84	5.36	5.57	5.88	4.88	3.56	2.05	1.10	0.88	(44)	
(45)		RadStd	0.13	0.19	0.31	0.43	0.38	0.46	0.35	0.38	0.32	0.28	0.14	0.10	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (19 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page